

Chapter 3 Revision Questions

SDS 188

2023-03-15

Question 1

Given the price of a share L at 8 periods in time:

Time	1	2	3	4	5	6	7	8
L	1720	1880	2780	2040	2060	2860	2230	1370

- (a) Calculate the mean ($\bar{x}$) share price of L .
- (b) Calculate the geometric mean ($\bar{x}_G$) share price of L .

	r_0	r_1	r_2	r_3	r_4	r_5	r_6	r_7
r_i	-	$(C_a)\%$	$(C_b)\%$	0.98%	38.83%	38.83%	-22.03%	-38.57%

- (c) Calculate the rate at which the investment grew in year 1 (C_a) and year 2 (C_b):
- (d) Calculate the geometric mean rate at which the investment grew each year ($\bar{r}_G$):

Question 2

Use the data from Question 1 to answer the following:

- (a) Determine the sample variance (s^2) of the price of share L :

$$s^2 = \sum_{i=1}^n \frac{1}{n-1} (x_i - \frac{1}{n} \sum_{j=1}^n x_j)^2$$

- (b) Calculate the standard deviation (s):
- (c) Calculate the Coefficient of Variation of the price of L :

Question 3

30 Different prices were collected for a single unit of commodity U . The prices are given in the form of a grouped data summary below:

Price Range	Frequency(f)	Midpoint(x)	$f \cdot x$	x^2	$f \cdot x^2$
1-20	2				
21-40	6				
41-60	7				
61-80	8				
81-100	4				
101-120	3				
total	n=30	-			

- (a) *Approximate the mean*
(b) *Approximate the variance*

$$s^2 = \sum_{i=1}^n \frac{1}{n-1} (x_i - \frac{1}{n} \sum_{j=1}^n x_j)^2$$

$$= \frac{1}{n-1} \left[\sum f \cdot x^2 - \frac{1}{n} \left(\sum f \cdot x \right)^2 \right]$$

Rewrite in alternative form

Question 4

- (a) *Given a RV X that follows a Normal distribution with a mean of 50 and a variance of 25, find the z-scores for the following three values:*

$$x_1 = 27.5 \qquad z_1 =$$

$$x_2 = 49 \qquad z_2 =$$

$$x_3 = 60 \qquad z_3 =$$

- (b) *Which one of these observations can be considered an outlier? Motivate.*

Question 5

You are given the following stem and leaf diagram:

1	3789
2	158999
3	0014
4	455
5	25
6	9

- (a) *Find the range*
(b) *Find the median*

Median position:

$$\frac{(n+1)}{2} =$$

(c) *Give the mode*

(d) *Comment on the shape of the distribution*

Question 6 (Example 3.13 in the prescribed textbook)

Nutritional data about a sample of seven breakfast cereals includes the number of calories per serving. Compute the five-number summary of the number of calories in cereals:

Cereal	Calories
Kellogg's All Bran	80
Kellogg's Corn Flakes	100
Wheaties	100
Nature's Path Organic Multigrain Flakes	110
Kellogg's Rice Krispies	130
Post Shredded Wheat Vanilla Almond	190
Kellogg's Mini Wheats	200

Min =

Max =

Q1 position:

$$\frac{(n+1)}{4} =$$

Median/Q2 position:

$$\frac{(n+1)}{2} =$$

Q3 position:

$$\frac{3(n+1)}{4} =$$

The five-number summary is:

Min	Q1	Q2	Q3	Max

Question 7

A population of 2-liter bottles of cola is known to have a mean fill-weight of 2.06 liters and a standard deviation of 0.02 liter.

- (a) The population is known to be bell-shaped. Describe the distribution of fill-weights. Is it very likely that a bottle will contain less than 2 liters of cola?

$$\mu \pm \sigma =$$

$$\mu \pm 2\sigma =$$

$$\mu \pm 3\sigma =$$

- (b) Suppose now that the shape of the population is unknown, and you cannot assume that it is bell-shaped. Describe the distribution of fill-weights. Is it very likely that a bottle will contain less than 2 liters of cola?

$$\mu \pm \sigma =$$

$$\mu \pm 2\sigma =$$

$$\mu \pm 3\sigma =$$

- (c) For some RV X that is not normally distributed, with mean 50 and variance of 25, find the bounds for the interval that should contain at least 95% of the data.

Given Chebyshev's theorem:

$$P(\mu - k\sigma \leq x \leq \mu + k\sigma) \geq \left(1 - \frac{1}{k^2}\right)$$

$$P(50 - k \cdot 5 \leq x \leq 50 + k \cdot 5) \geq \left(1 - \frac{1}{k^2}\right)$$

Find the value for k (round final answer to 2 decimals):

$$\left(1 - \frac{1}{k^2}\right) = 0.95$$

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Question 8

A sample of size 10 is collected on two variables, X and Y :

X	16	18	18	19	21	14	18	26	17	21
Y	47	48	57	50	50	58	52	43	46	47

(a) Calculate the covariance

$$\begin{aligned} \text{cov}(X, Y) &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i y_i - \frac{1}{n} \left[\sum_{i=1}^n x_i \sum_{i=1}^n y_i \right] \right) \\ &= \end{aligned}$$

(b) Give a disadvantage of the covariance

(c) Calculate the correlation

$$\begin{aligned} r &= \frac{\text{cov}(X, Y)}{s_x s_y} \\ &= \end{aligned}$$